QUAID-I-AZAM UNIVERSITY ISLAMABAD

DEPARTMENT OF COMPUTER SCIENCE



DETECTION OF CLICKBAIT THROUGH MACHINE LEARNING

SOFTWARE PROJECT TECHNICAL DOCUMENTATION

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Submitted to:

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Abstract

Detection of clickbait through Machine Learning

Due to the fast development of social media and information websites, misleading and exaggerated news titles, have become increasingly more commonplace. These misleading information titles goal to seize interest of the reader and maximize the quantity of clicks. However, they'll mislead the reader and create false expectations. In this assignment, the development of a **Clickbait Detection System** based totally on system getting to know may be discussed. The system might be capable to analyze news titles or small information articles to decide whether they're clickbait or valid information.

To increase the **Clickbait Detection System, Natural Language Processing (NLP)** strategies might be hired to procedure the input information, extract essential features, and train a class version to become aware of clickbait, The system might be capable of be given news titles or small news articles as input and offer an immediate prediction with a self assurance level. The machine will also be capable of pick out the vital phrases or terms that inspired the prediction to help the person better recognize the purpose behind the prediction. In addition to this , the system can also be able to maintain a history of the predictions and offer accuracy, precision, and take into account metrices to determine the performance of the type version.

By integrating gadget mastering with the prediction mechanisms, the proposed Clickbait Detection System may be an sensible machine to become aware of misleading news titles. The proposed device could be beneficial for customers to critically examine the virtual content they eat, and the overall recognition of deceptive news titles can be elevated

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Chapter No. 01

Software Project Management Plan

3.1.3 Implementation Phase

3.1.3.1 System Implementation

Task Description	This phase makes a speciality of the improvement and integration of the clickbait detection device with the aid of applying device gaining knowledge of techniques, textual content evaluation techniques, and machine additives to provide prediction and rationalization effects.	
Deliverables	After completing the software requirements, use case details and diagrams will be created.	
Resources Needed	**People**	Faizan Ahmad, Zohaib Zafar
	Software	Microsoft Word, Python, Machine Learning libraries, Web development Tools
	Hardware	Laptop
Dependencies and Constraints	We have prepared clickbait statistics and trained models that may be used in the mission.	
Risks	Model consequences depend upon the exceptional and balance of the schooling facts.	

3.2 Assignments

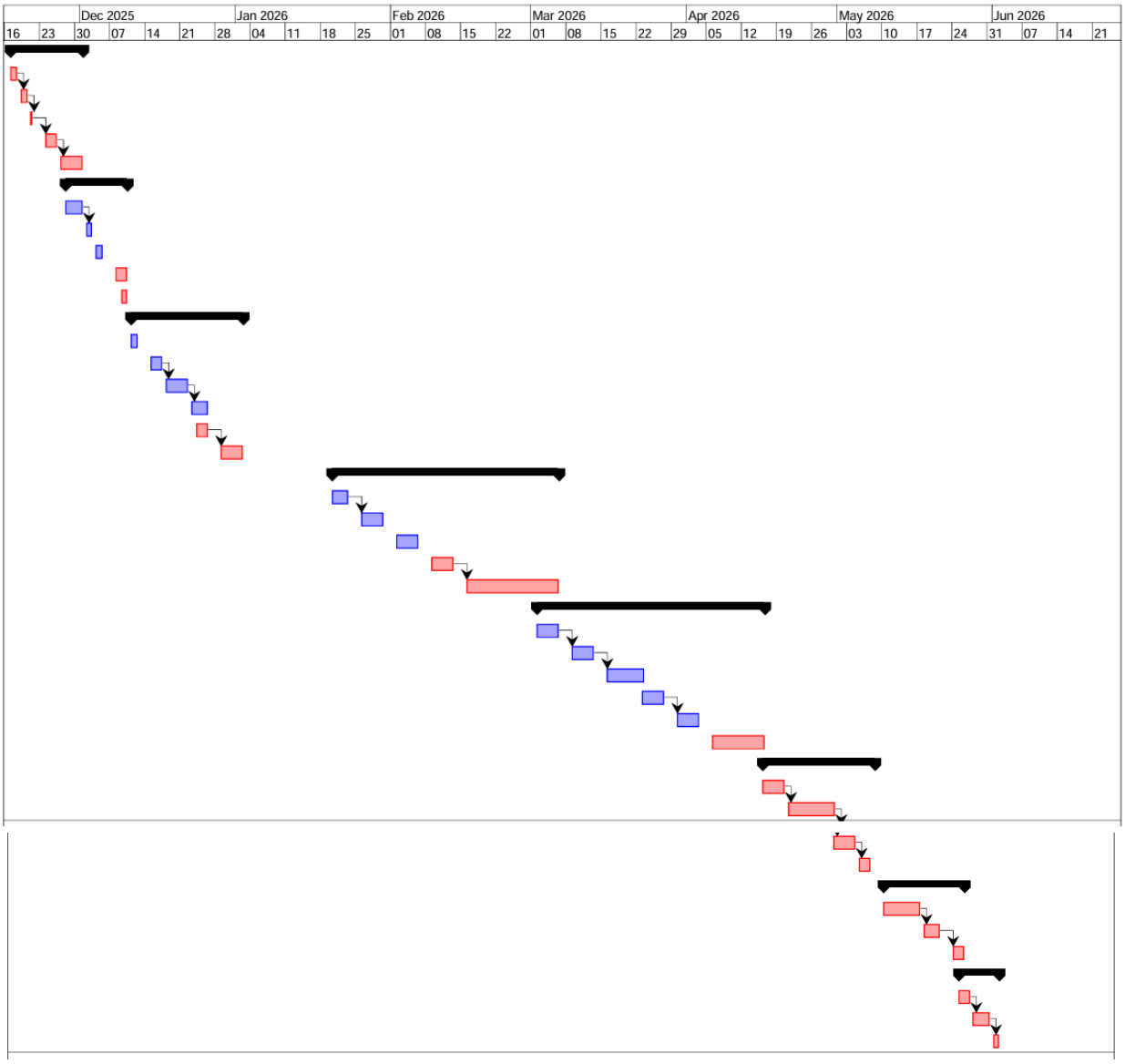
Task	Assignee	Role
Requirement identification	FAIZAN AHMAD, ZOHAIB ZAFAR	SYSTEM Analyst
Defining Use Cases		SYSTEM Analyst
Develop Domain Model		SYSTEM Analyst, Software Architect.
ERD Design		Database Designer, SYSTEM Analyst
Testing and Validation		Performing device testing to confirm features and outcomes.
Documentation		Creating complete technical documentation for the very last year assignment.
User Interface Design		Designing how customers engage with the system and notice consequences.
Class Diagram Designing		Software Architect, SYSTEM Analyst.

Sequence Diagrams		SYSTEM Analyst, Software Architect
Design Phase Verification		Quality Assurance Engineer, SYSTEM Software Architect.
Software Test Documentation		Quality Assurance Engineer, Test Analyst
Implementation Phase		Software Developer.
Review and Evaluation		Reviewing challenge development, validating outcomes, and providing instructional steerage.

3.3 Project Schedule

		Name	Duration	Start	Finish	Predecessors	Resource Names
1		**Project Initiation & Pla...**	**11 days?**	**11/17/25, 8:00AM**	**12/1/25, 5:00PM**		
2		Topic finalization	2 days?	11/17/25, 8:00AM	11/18/25, 5:00PM		
3		Problem statement defini...	2 days?	11/19/25, 8:00AM	11/20/25, 5:00PM	2	
4		Literature review (existi...	1 day?	11/21/25, 8:00AM	11/21/25, 5:00PM	3	
5		Project scope & objectiv...	3 days?	11/24/25, 8:00AM	11/26/25, 5:00PM	4	
6		Initial project plan approva	3 days?	11/27/25, 8:00AM	12/1/25, 5:00PM	5	
7		**Requirement Analysis**	**9 days?**	**11/28/25, 8:00AM**	**12/10/25, 5:00PM**		
8		Functional requirements ...	2 days?	11/28/25, 8:00AM	12/1/25, 5:00PM		
9		Non-functional requirem...	2 days?	12/2/25, 8:00AM	12/3/25, 5:00PM	8	
10		Use case identification (...	2 days?	12/4/25, 8:00AM	12/5/25, 5:00PM		
11		Requirements validation	3 days?	12/7/25, 8:00AM	12/10/25, 5:00PM		
12		Requirements document...	2 days?	12/9/25, 8:00AM	12/10/25, 5:00PM		
13		**System Design**	**17 days?**	**12/11/25, 8:00AM**	**1/2/26, 5:00PM**		
14		High-level system archit...	2 days?	12/11/25, 8:00AM	12/12/25, 5:00PM		
15		Use Case Diagram design	3 days?	12/13/25, 8:00AM	12/17/25, 5:00PM		
16		System Sequence Diagra...	3 days?	12/18/25, 8:00AM	12/22/25, 5:00PM	15	
17		Sequence Diagrams (UC-...	4 days?	12/23/25, 8:00AM	12/26/25, 5:00PM	16	
18		Database & data flow de...	3 days?	12/24/25, 8:00AM	12/26/25, 5:00PM		
19		Design review & refinem...	5 days?	12/29/25, 8:00AM	1/2/26, 5:00PM	18	
20		**Model Development**	**34 days?**	**1/20/26, 8:00AM**	**3/6/26, 5:00PM**		
21		Dataset collection	4 days?	1/20/26, 8:00AM	1/23/26, 5:00PM		
22		Data cleaning & preproc...	5 days?	1/26/26, 8:00AM	1/30/26, 5:00PM	21	
23		Feature extraction & sel...	5 days?	2/2/26, 8:00AM	2/6/26, 5:00PM		
24		Machine learning model s..	5 days?	2/9/26, 8:00AM	2/13/26, 5:00PM		
25		Model training & evaluatior	15 days?	2/16/26, 8:00AM	3/6/26, 5:00PM	24	
26		**System Development**	**34 days?**	**3/2/26, 8:00AM**	**4/16/26, 5:00PM**		
27		User authentication module	5 days?	3/2/26, 8:00AM	3/6/26, 5:00PM		
28		Headline input & predicti...	5 days?	3/9/26, 8:00AM	3/13/26, 5:00PM	27	
29		Explanation generation ...	6 days?	3/16/26, 8:00AM	3/23/26, 5:00PM	28	
30		Detection history manag...	5 days?	3/21/26, 8:00AM	3/27/26, 5:00PM		
31		Model performance visu...	5 days?	3/30/26, 8:00AM	4/3/26, 5:00PM	30	
32		system integration	9 days?	4/4/26, 8:00AM	4/16/26, 5:00PM		
33		**Testing & Validation**	**17 days?**	**4/16/26, 8:00AM**	**5/8/26, 5:00PM**		
34		Unit testing	3 days?	4/16/26, 8:00AM	4/20/26, 5:00PM		
35		Functional testing	8 days?	4/21/26, 8:00AM	4/30/26, 5:00PM	34	
36		Acceptance testing	3 days?	5/1/26, 8:00AM	5/5/26, 5:00PM	35	
37		Bug fixing & validation	3 days?	5/6/26, 8:00AM	5/8/26, 5:00PM	36	
38		**Documentation**	**13 days?**	**5/11/26, 8:00AM**	**5/27/26, 5:00PM**		
39		Technical documentation...	6 days?	5/11/26, 8:00AM	5/18/26, 5:00PM		
40		Diagram & table updates	4 days?	5/19/26, 8:00AM	5/22/26, 5:00PM	39	
41		Proofreading & formatting	3 days?	5/25/26, 8:00AM	5/27/26, 5:00PM	40	
42		**Final Review & Submis...**	**7 days?**	**5/26/26, 8:00AM**	**6/3/26, 5:00PM**		
43		Supervisor review	3 days?	5/26/26, 8:00AM	5/28/26, 5:00PM		
44		Final corrections	2 days?	5/29/26, 8:00AM	6/1/26, 5:00PM	43	
45		Final submission	2 days?	6/2/26, 8:00AM	6/3/26, 5:00PM	44	

3.4 Gantt Chart



Chapter 2

Software Requirement Specification

2.1 Introduction

This document describes the expected software program features, constraints, interfaces and different attributes. In this record we can define instances, sequence diagrams, area model and database layout of the venture.

2.2 Functional Requirements

The functional requirements for the Clickbait Detection Through Machine gaining knowledge of System are as fellow:

1. The system need to permit users to create an account with the aid of presenting valid registration statistics.

2. The device ought to allow registered users to authenticate themselves the usage of stable login credentials.

3. The machine should offer a password recovery mechanism in case customers forget about their credentials.

4. The device ought to permit users to terminate their active session securely.

5. The machine ought to permit users to submit news headlines or article textual content for evaluation.

6. The system have to assist textual content input in English language with varying headline lengths.

7. The system need to preprocess the submitted text by using:

   A. Removing needless symbols and punctuation.

   B. Converting text into a standardized layout.

   C. Handling stop phrases and redundant phrases.

8. The system need to classify the submitted headline or textual content into:

   a. Clickbait

   b. Non-Clickbait

9. The machine ought to calculate a self belief score indicating the chance of the content material being clickbait.

10. The device ought to gift the prediction end result honestly to the person.11. The unit shall provide predictive interpretation records through identifying influential words or terms that make contributions to elegance choice.

12. The unit shall spotlight linguistic styles commonly related to clickbait, which includes:

A. Sensational words

B. Emotional trigger

C. Unclear Terms

D. Hyperbolic expression

13. The machine shall permit customers to upload statistics units in CSV format for analytical purposes.

14. The device have to validate the uploaded records set the usage of checks:

A. Record layout

B. Data consistency

C. Missing or Corrupt Entries

15. The system need to allow legal clients to provoke gaining knowledge of strategies the use of the uploaded datasets.

16. The Gadget shall calculate and display overall performance metrics, which includes:

A. Accuracy

B. Precision

C. Remember

D. F1 score

17. The unit ought to assemble the very last result of every detection:

A. Submitted text

B. Prediction of end end result

C. Confidence assessment

D. Date and time of evaluation

18. The machine must allow users to view formerly analyzed headlines and results.

19. The system must allow customers to go looking and filter detection history based totally on date or prediction kind.

20. The machine must make sure that saved facts stays steady and unaltered throughout sessions.

21. The machine must restriction access to superior analytical features primarily based on person role.

22. The system should ensure that prediction results are generated within an acceptable response time.

23. The system should handle invalid or incomplete input gracefully without system failure.

24. The system should maintain activity logs for analytical and auditing purposes.

Use Case Diagram

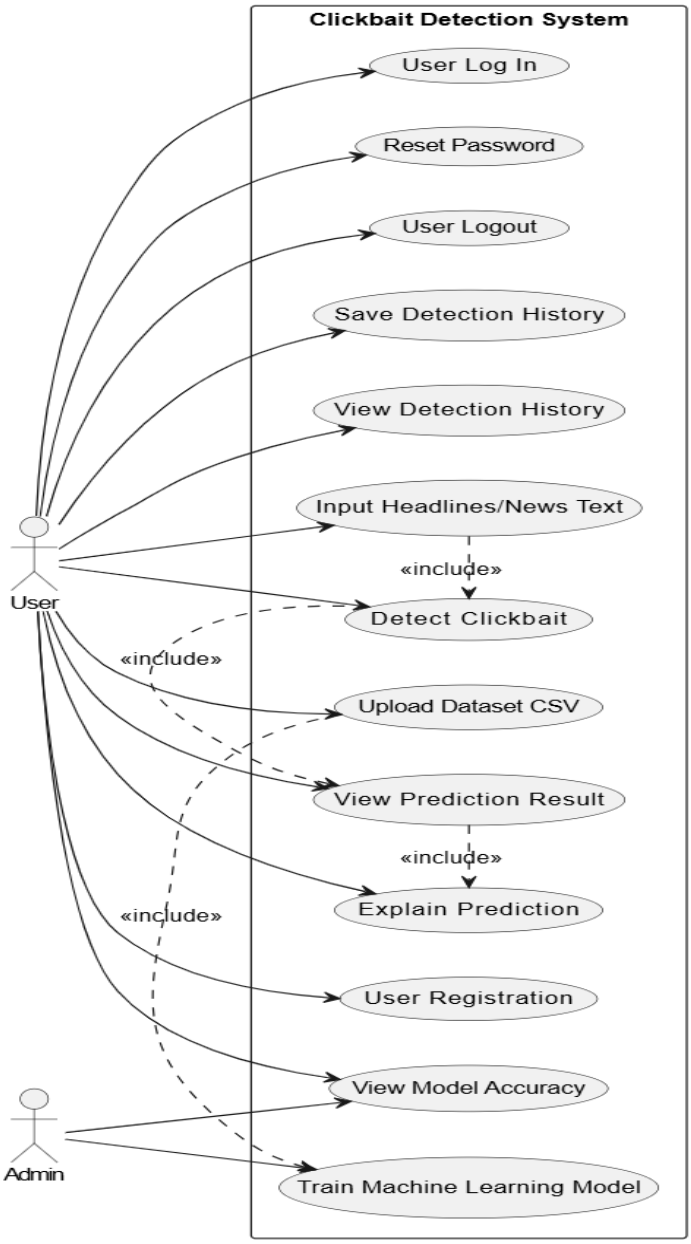


Fig 2 Use Case Diagram

Use Case Description

UC-1 User Registration

USE CASE ID	UC-1
USE CASE NAME	User Registration
Actors	User
Description	It describes how a brand new consumer signs and symptoms up through getting into required statistics.
Preconditions	1. User has access to the Clickbait Detection System. 2. User is not already registered in the system.
Postconditions	A new user account is successfully created. User information is securely stored in the SYSTEM.
Normal Flow	1. User opens the registration interface. 2. User enters required details along with name, electronic mail, and password. 3. User submits the registration shape. 4. The gadget validates the entered information. 5. The device creates a brand new user account. 6. The device displays a a success registration confirmation.
Alternative Flow	**3a.** If any required field is missing, the system prompts the user to complete the missing information. **3b.** If the entered email already exists, the system notifies the user to use a different email address
Exception	1. Invalid input format. 2. System failure during the registration process.
Special Requirements	1. User credentials need to be saved in encrypted shape. 2. Email addresses need to be unique across the system. 3. The registration manner have to whole within perfect reaction time.
Frequency of Use	Low
Technology and Data variation List	1. Registration is performed thru a web-based totally interface. 2. User input records can also vary in period and layout. 3. The device should assist multiple customers registering concurrently.

Table 1 Use Case Description UC-1 User Registration

UC-2 User Login

USE CASE ID	UC-2
USE CASE NAME	User Login
Actors	User

Description	It describes how a consumer signs in the usage of a legitimate username and password.
Preconditions	1. User must be registered. 2. User must have valid login credentials
Postconditions	1. User is efficiently authenticated. 2. User is redirected to the system dashboard.
Normal Flow	1. User opens login page. 2. User enters email and password. 3. User submits login request. 4. System verifies credentials. 5. System grants access.
Alternative Flow	**3a.** If credentials are incorrect, SYSTEM displays an error message. **3b.** If fields are empty, SYSTEM prompts user to complete them.
Exception	System authentication failure.
Special Requirements	Secure authentication mechanism must be used.
Frequency of Use	High
Technology and Data variation List	Web-based login, variable credential formats.

Table 2 Use Case Description UC-2 User Login

UC-3 Reset Password

USE CASE ID	UC-3
USE CASE NAME	Reset Password
Actors	User
Description	It shows the technique of password recuperation for registered users.
Preconditions	User must be registered.
Postconditions	Password is successfully updated.
Normal Flow	1. User selects reset password choice. 2. User enters registered e mail. 3. System verifies electronic mail. 4. User units new password. 5. System updates password.
Alternative Flow	**2a.** Email not found → SYSTEM shows error.
Exception	SYSTEM error during password update.
Special Requirements	Password reset must follow security rules.
Frequency of Use	Low

Technology and Data variation List	Email-based verification, varied password formats.

Table 3 Use Case Description UC-3 Reset Password

UC-4 User Logout

USE CASE ID	UC-4
USE CASE NAME	User Logout
Actors	User
Description	It describes how a person symptoms out of the system.
Preconditions	User must be logged in.
Postconditions	User session is terminated.
Normal Flow	1. User selects logout choice. 2. System ends consultation. 3. User is redirected to login web page.
Alternative Flow	N/A
Exception	Session termination failure.
Special Requirements	Session termination failure.
Frequency of Use	Medium.
Technology and Data variation List	Session-based authentication handling.

Table 4 Use Case Description UC-4 User Logout

UC-5 Input Headlines / News Text

USE CASE ID	UC-5
USE CASE NAME	Input Headlines / News Text
Actors	User
Description	This use case permits customers to analyze headlines or news textual content.
Preconditions	User must be logged in.
Postconditions	Text is submitted for processing.
Normal Flow	1. User opens text input interface. 2. User enters headline or news text. 3. User submits text.
Alternative Flow	**2a.** Empty input → system prompts user.
Exception	Invalid text format.
Special Requirements	Input length limits must be enforced.
Frequency of Use	High

Technology and Data variation List	Text length and language variation.

Table 5 Use Case Description UC-5 Input Headlines / News Text

UC-6 Upload Dataset

USE CASE ID	UC-6
USE CASE NAME	Upload Dataset
Actors	User
Description	This use case explains how customers add statistics to improve the model.
Preconditions	User must be logged in.
Postconditions	Dataset is stored in the system.
Normal Flow	1. User selects add dataset option. 2. User uploads CSV report. 3. System validates report layout. 4. Dataset is saved.
Alternative Flow	**2a.** Invalid file → system rejects upload.
Exception	File processing error.
Special Requirements	Only CSV format allowed.
Frequency of Use	Low
Technology and Data variation List	Only CSV format allowed.

Table 6 Use Case Description UC-6 Upload Dataset

UC-7 Train Machine Learning Model

USE CASE ID	UC-7
USE CASE NAME	Train Machine Learning Model
Actors	User
Description	It describes how the system trains the model the use of existing facts.
Preconditions	Dataset must be available.
Postconditions	Model is skilled and geared up for use.
Normal Flow	1. Admin selects training option. 2. System approaches dataset. 3. Model is educated. 4. Training finishing touch is showed.
Alternative Flow	**2a.** Insufficient data → training aborted.
Exception	Model training failure.

Special Requirements	Training consequences need to be saved.
Frequency of Use	Low
Technology and Data variation List	Different dataset sizes and features.

Table 7 Use Case Description UC-7 Train Machine Learning Model

UC-8 Detect Clickbait

USE CASE ID	UC-8
USE CASE NAME	Detect Clickbait
Actors	User
Description	It explains how the machine classifies textual content as clickbait or non-clickbait.
Preconditions	Text must be provided.
Postconditions	Prediction result is generated.
Normal Flow	1. User submits headline or information textual content for analysis. 2. The system completes the detection process. 3. The system shows the prediction result to the user.
Alternative Flow	**1a.** Empty input → prompt user.
Exception	Prediction processing error.
Special Requirements	Prediction response time must be reasonable.
Frequency of Use	High.
Technology and Data variation List	Different writing styles and vocabulary.

Table 8 Use Case Description UC-8 Detect Clickbait

UC-9 View Prediction Result

USE CASE ID	UC-9
USE CASE NAME	View Prediction Result
Actors	User
Description	This use case suggests how the device shows effects after textual content submission.
Preconditions	1. User must be logged in. 2. A clickbait detection process must be completed.
Postconditions	Prediction Result is display to the user. The person can understand whether or not the content is classified as clickbait or non-clickbait.

Normal Flow	1. User submits headline or information textual content for analysis. 2. The system completes the detection process. 3. The system shows the prediction result to the user.
Alternative Flow	**2a.** If prediction result is not available, the system informs the user to retry the detection process.
Exception	1. System error during result retrieval. 2. Prediction output failure.
Special Requirements	1. Prediction result must be displayed in a clear and understandable format. 2. Result need to be shown within a suitable response time.
Frequency of Use	High.
Technology and Data variation List	1. Result show may additionally range based on textual content length and category self belief. 2. Different output codecs may be used for result presentation.

Table 9 Use Case Description UC-9 View Prediction Result

UC-10 Explain Prediction

USE CASE ID	UC-10
USE CASE NAME	Explain Prediction
Actors	User
Description	This use case explains why the gadget categorized the textual content as clickbait or no longer.
Preconditions	Prediction must be available.
Postconditions	Explanation is displayed.
Normal Flow	1. User requests rationalization. 2. System shows influential elements.
Alternative Flow	**N/A**
Exception	Explanation generation failure.
Special Requirements	Explanation must be understandable to users.
Frequency of Use	Medium.
Technology and Data variation List	Variation in rationalization element degree.

Table 10 Use Case Description UC-10 Explain Prediction

UC-11 View Model Accuracy and Performance

USE CASE ID	UC-11
USE CASE NAME	Explain Prediction

Actors	User
Description	This use case explains why the machine classified the textual content as clickbait or no longer.
Preconditions	Prediction must be available.
Postconditions	Explanation is displayed.
Normal Flow	1. User requests clarification. 2. System presentations influential factors.
Alternative Flow	**N/A**
Exception	Explanation generation failure.
Special Requirements	Explanation need to be understandable to customers.
Frequency of Use	Medium.
Technology and Data variation List	Variation in explanation detail level.

Table 11 Use Case Description UC-11 View Model Accuracy and Performance

UC-12 Save Detection History

USE CASE ID	UC-12
USE CASE NAME	Save Detection History
Actors	User
Description	This use case saves prediction outcomes for destiny reference.
Preconditions	Prediction must exist.
Postconditions	Detection record is stored.
Normal Flow	1. System saves result automatically.
Alternative Flow	**N/A**
Exception	Storage failure.
Special Requirements	Data integrity should be maintained.
Frequency of Use	High.
Technology and Data variation List	Storage size variation.

Table 12 Use Case Description UC-12 Save Detection History

UC-13 View Detection History

USE CASE ID	UC-13
USE CASE NAME	View Detection History
Actors	User

Description	This use case lets in customers to look previous clickbait detection outcomes.
Preconditions	User must be logged in.
Postconditions	Detection history is displayed.
Normal Flow	1. User opens history page. 2. System retrieves records. 3. History is shown.
Alternative Flow	**2a.** No records → system shows empty state.
Exception	Retrieval error.
Special Requirements	History need to be looked after by way of date.
Frequency of Use	Medium.
Technology and Data variation List	Variable record count.

Table 13 Use Case Description UC-13 View Detection History

System Sequence Diagram

SYSTEM Sequence Diagrams Illustrate the interactions between outside actors and the SYSTEM displaying the waft of events and information over time.

SSD UC-1 User Registration

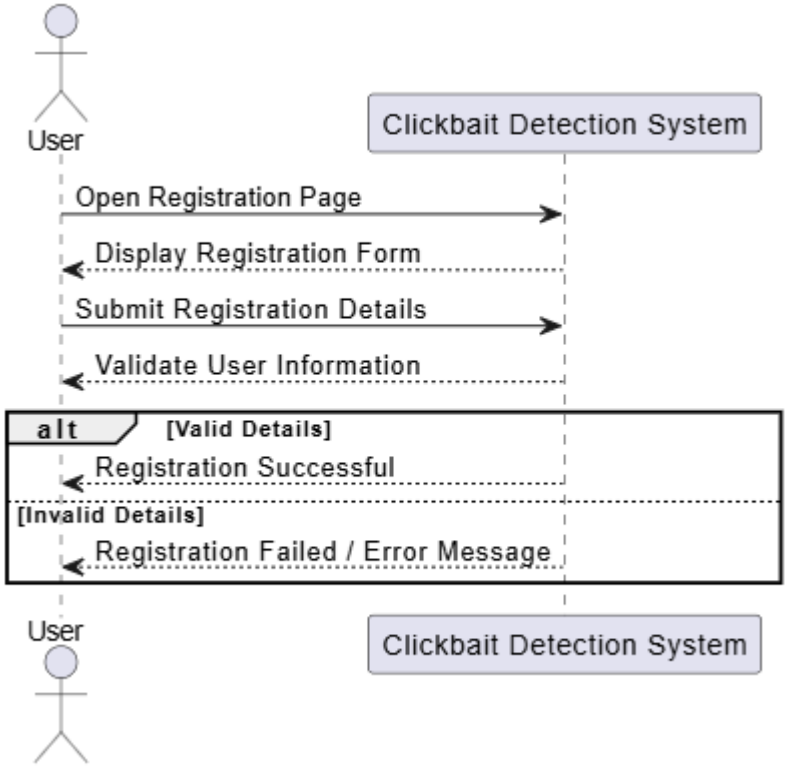


Figure 3 SSD UC-1 User Registration

SSD UC-2 User Login

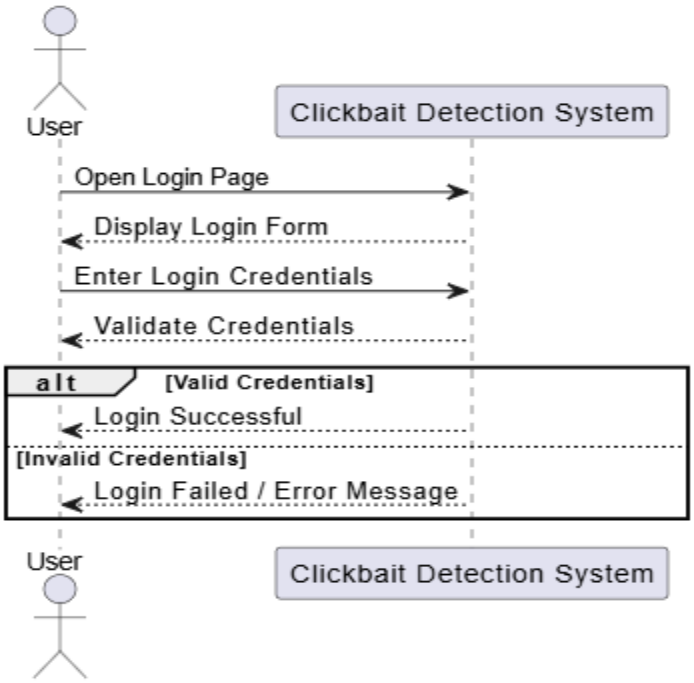


Figure 4 SSD UC-2 User Login

SSD UC-3 Reset Password

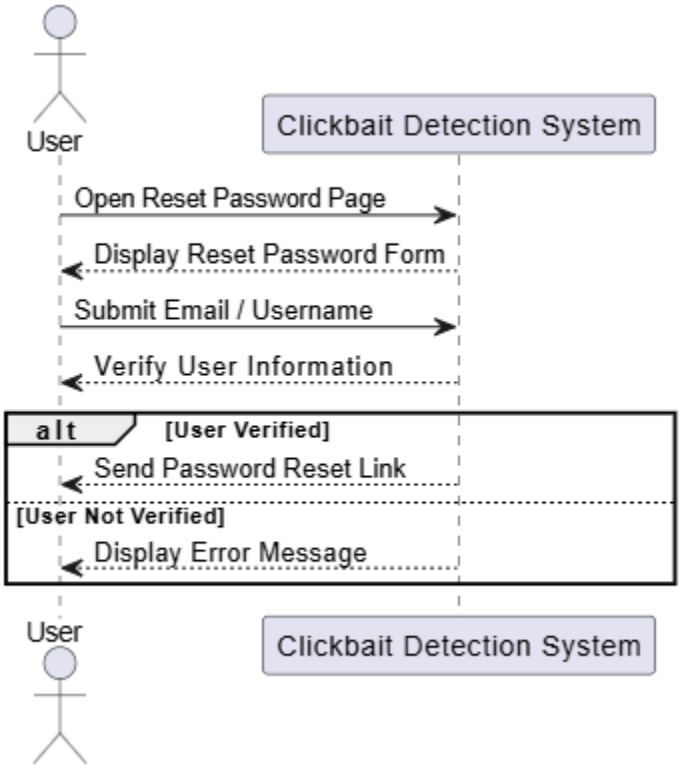


Figure 5 SSD UC-3 Reset Password

SSD UC-4 User Logout

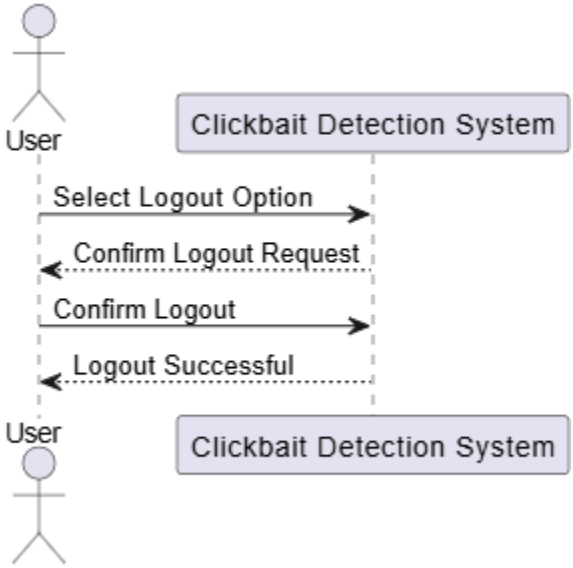


Figure 6 SSD UC-4 User Logout

SSD UC-5 Input Headlines / Text

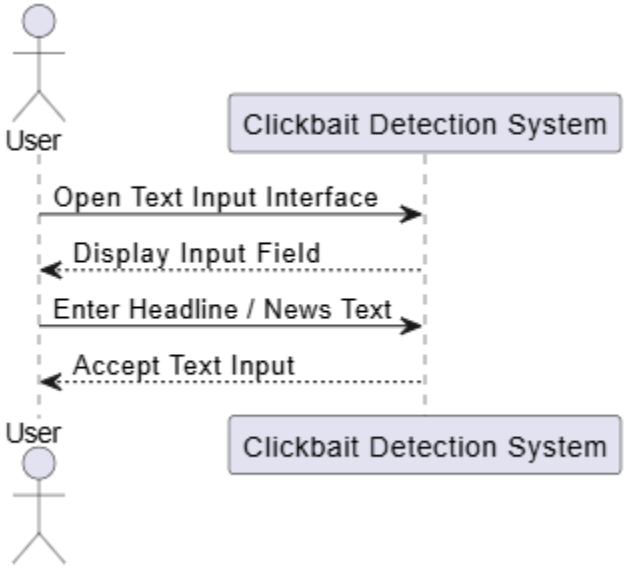


Figure 7 SSD UC-5 Input Headlines / Text

SSD UC-6 Upload Dataset

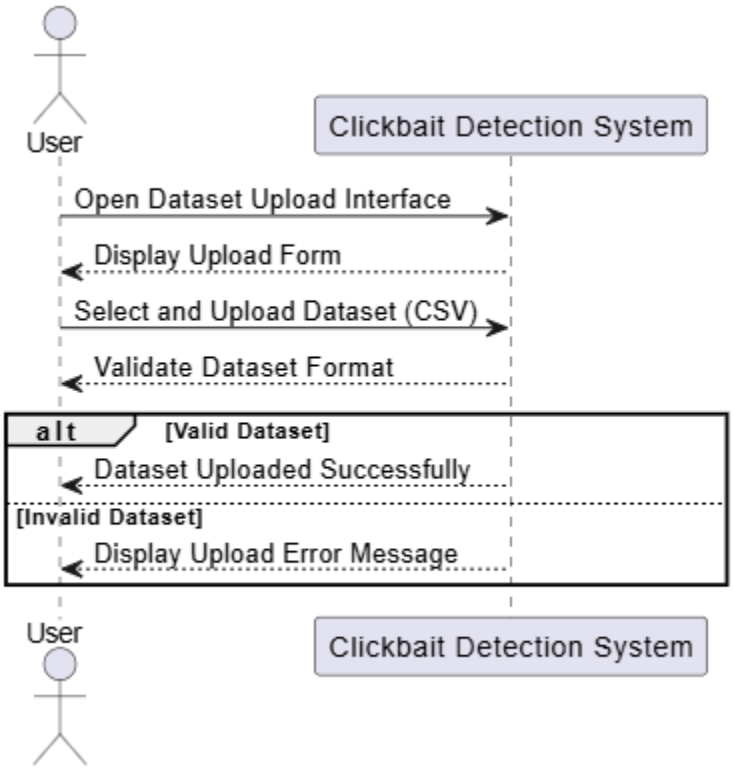


Figure 8 SSD UC-6 Upload Dataset

SSD UC-7 Train Machine Learning Model

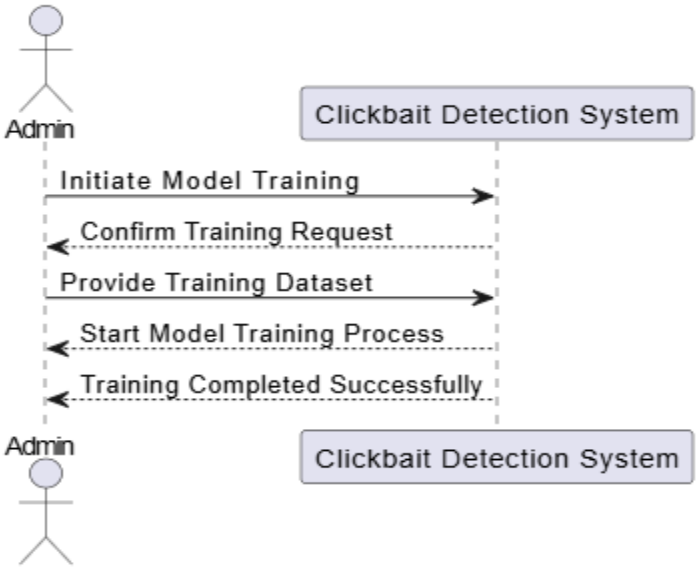


Figure 9 SSD UC-7 Train Machine Learning Model

SSD UC-8 Detect Clickbait

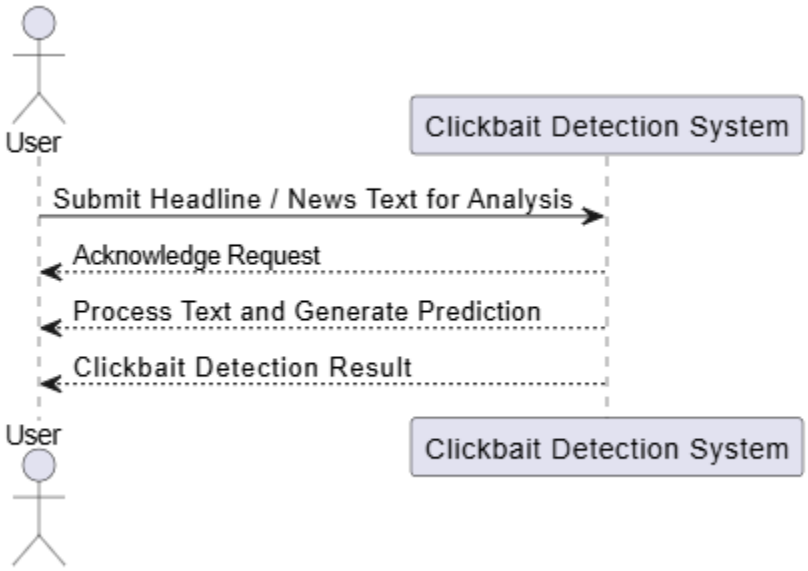


Figure 10 SSD UC-8 Detect Clickbait

SSD UC-9 Explain Prediction

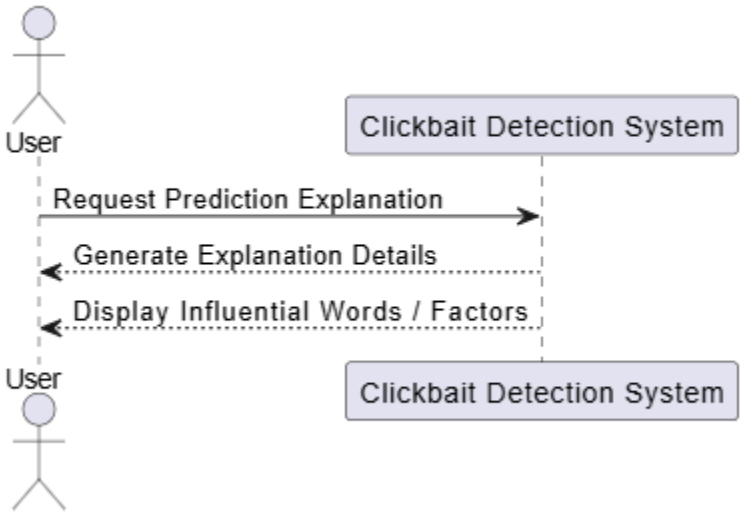


Figure 11 SSD UC-9 Explain Prediction

SSD UC-10 Save Detection History

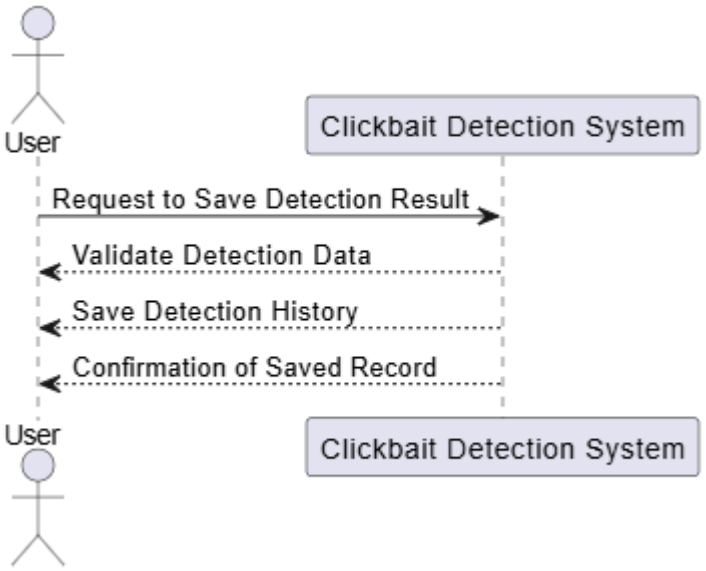


Figure 12 SSD UC-10 Save Detection History

SSD UC-11 View Detection History

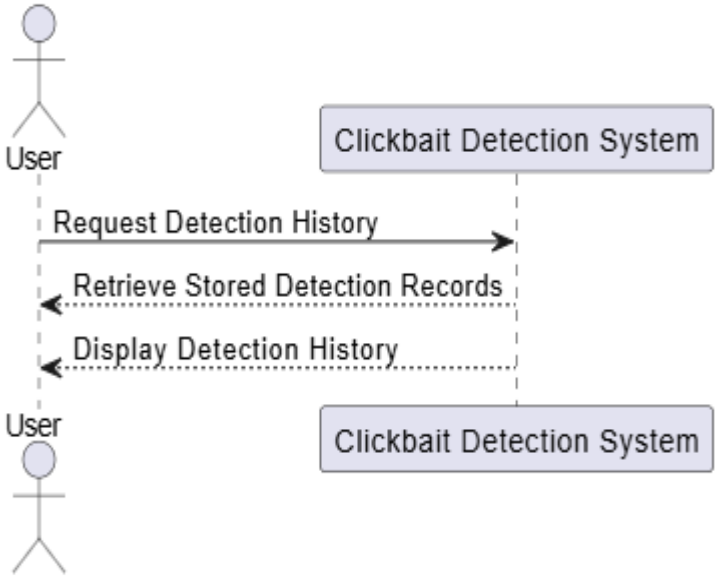


Figure 13 SSD UC-11 View Detection History

Domain Model

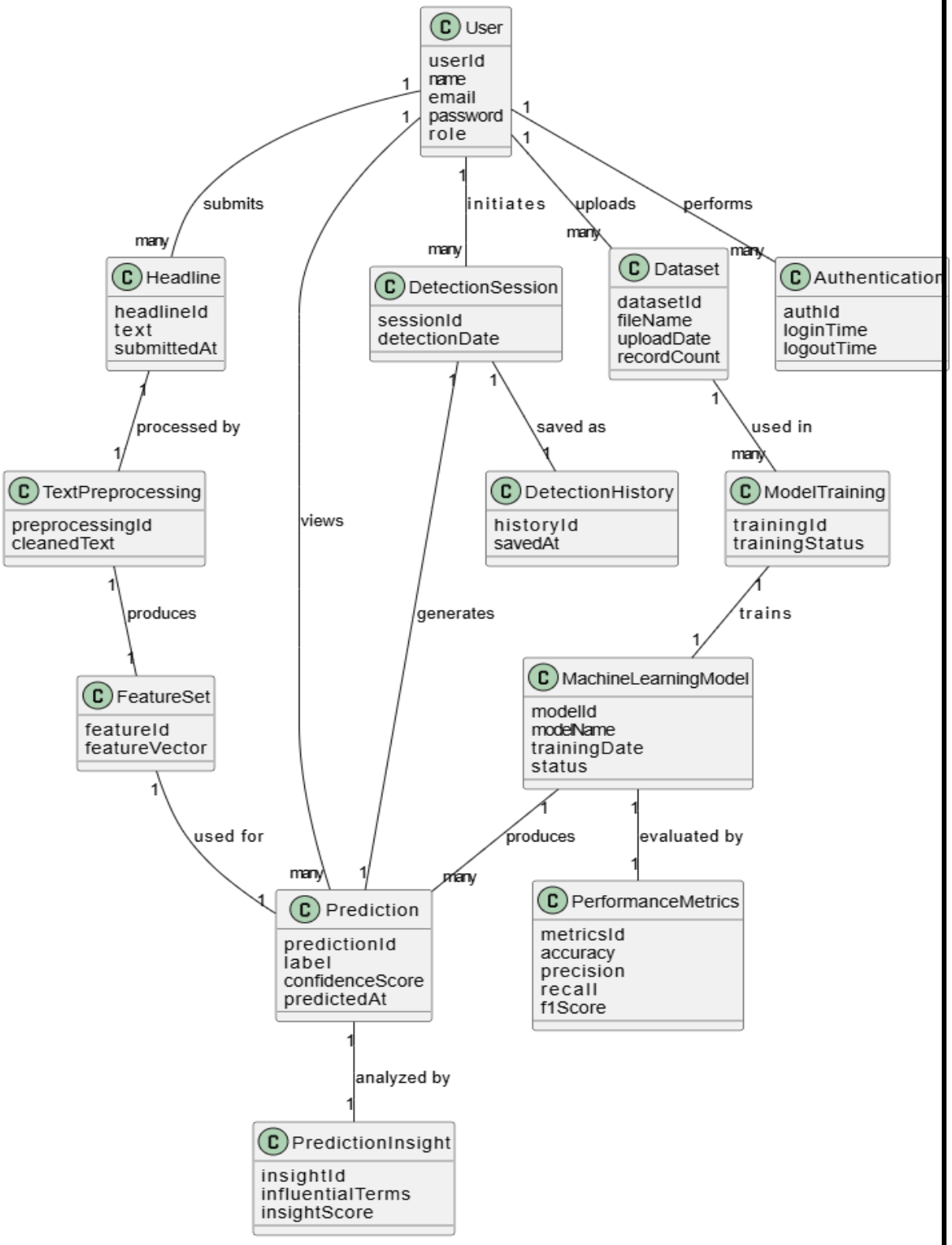


Figure 14 Domain Model

Chapter 3

Software Design and Description

Sequence Diagrams

SD UC-1 User Registration

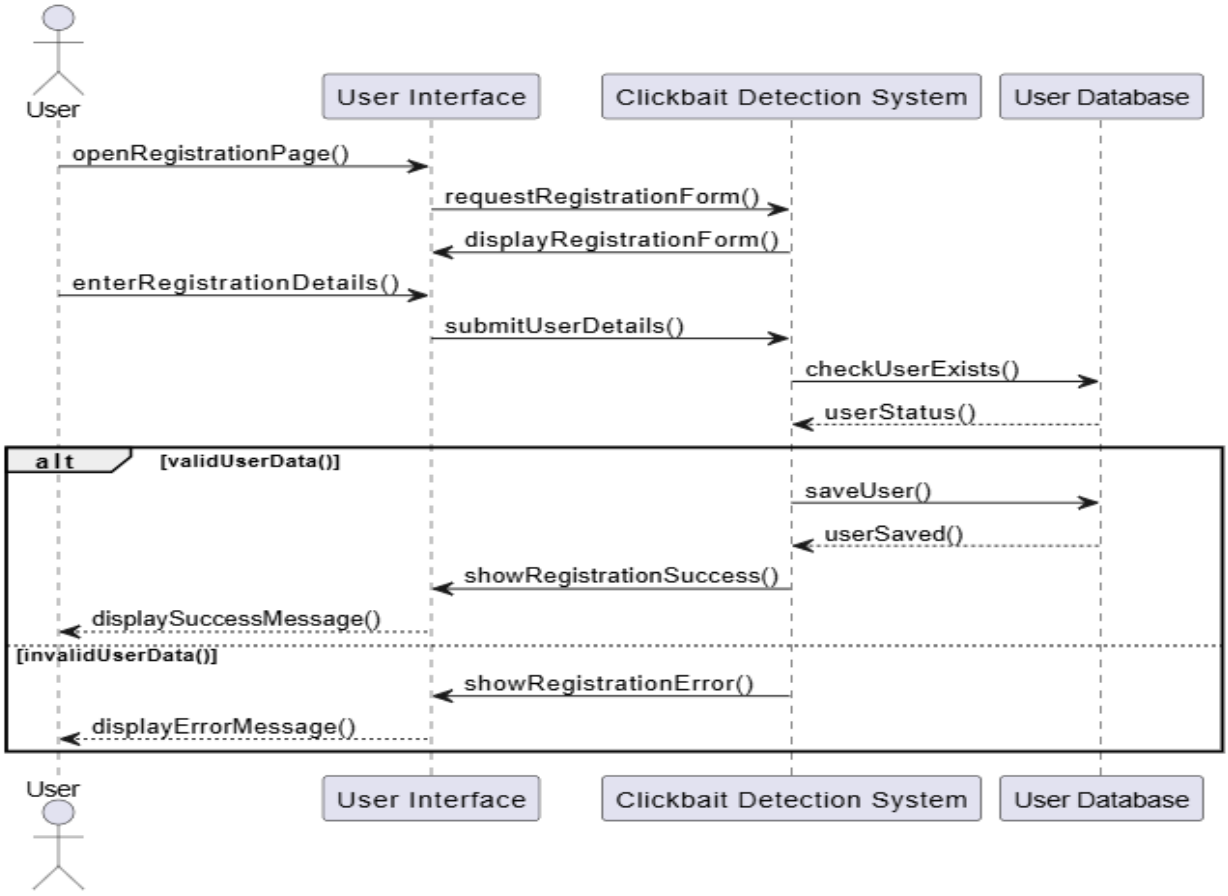


Figure 15 SD UC-1 User Registration

SD UC-2 User Login

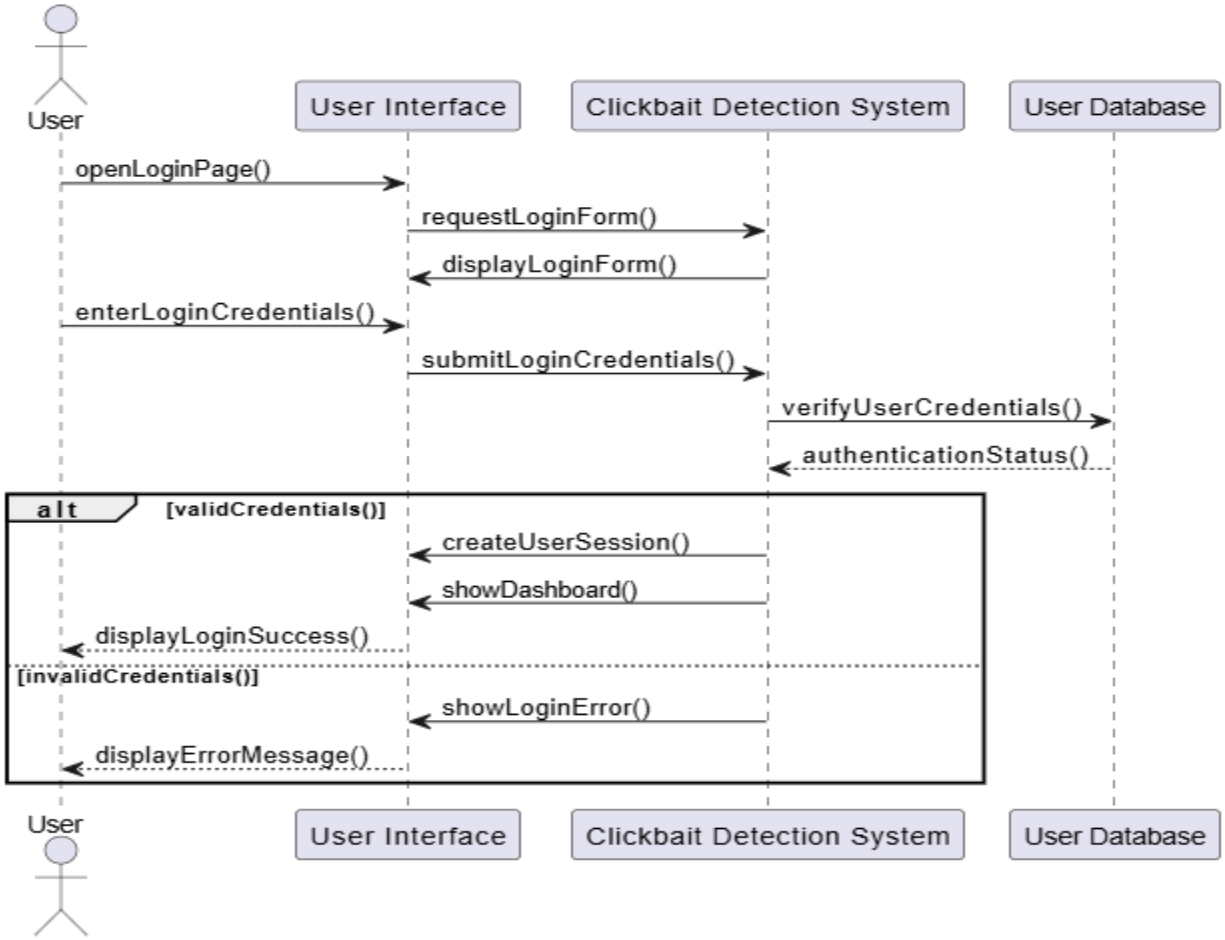


Figure 16 SD UC-2 User Login

SD UC-3 Reset Password

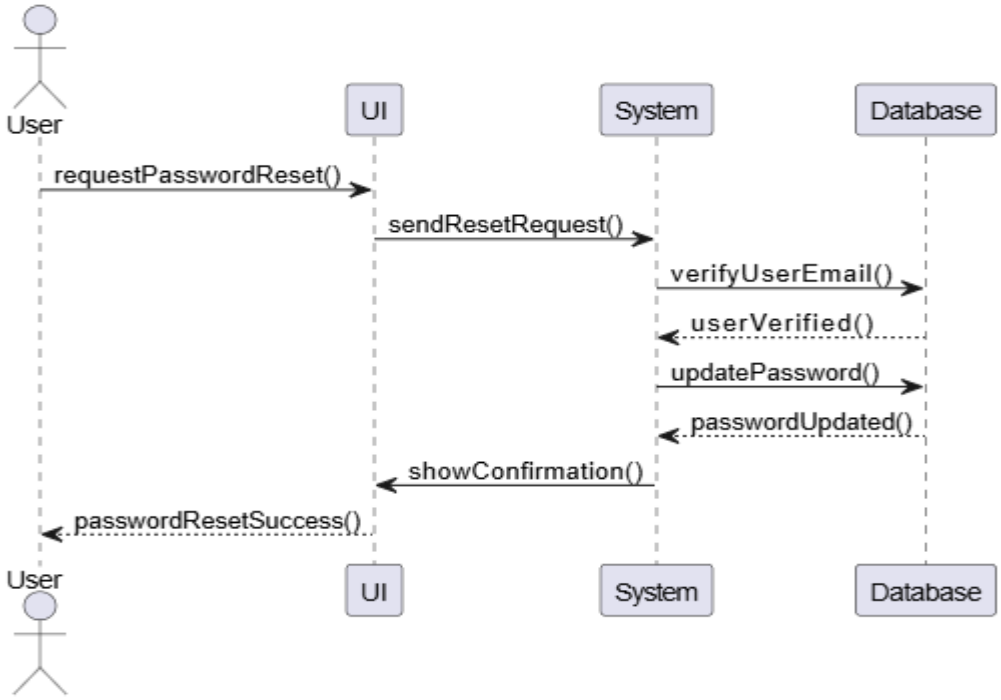


Figure 17 SD UC-3 Reset Password

SD UC-4 Input Headline / News text

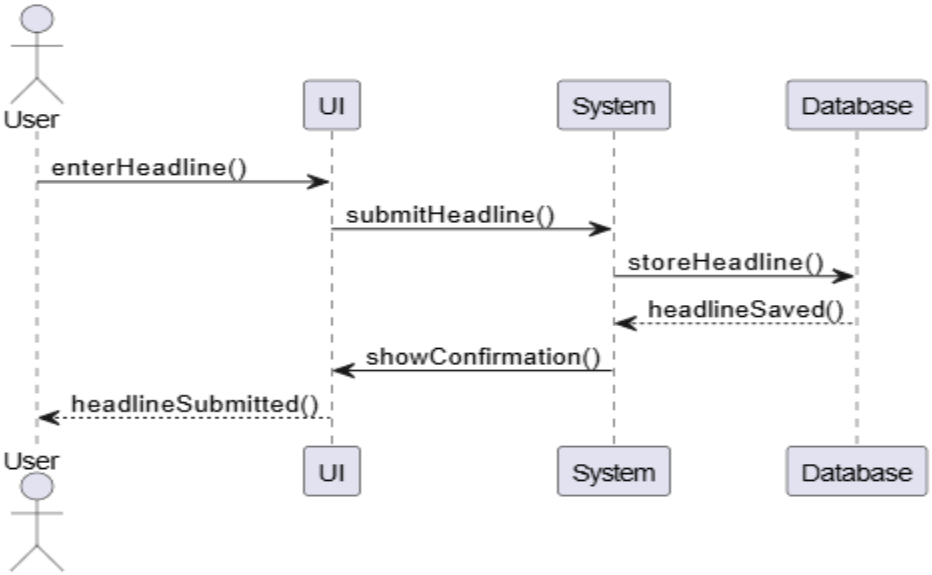


Figure 18 Input Headline / News text

SD UC-5 Text Preprocessing

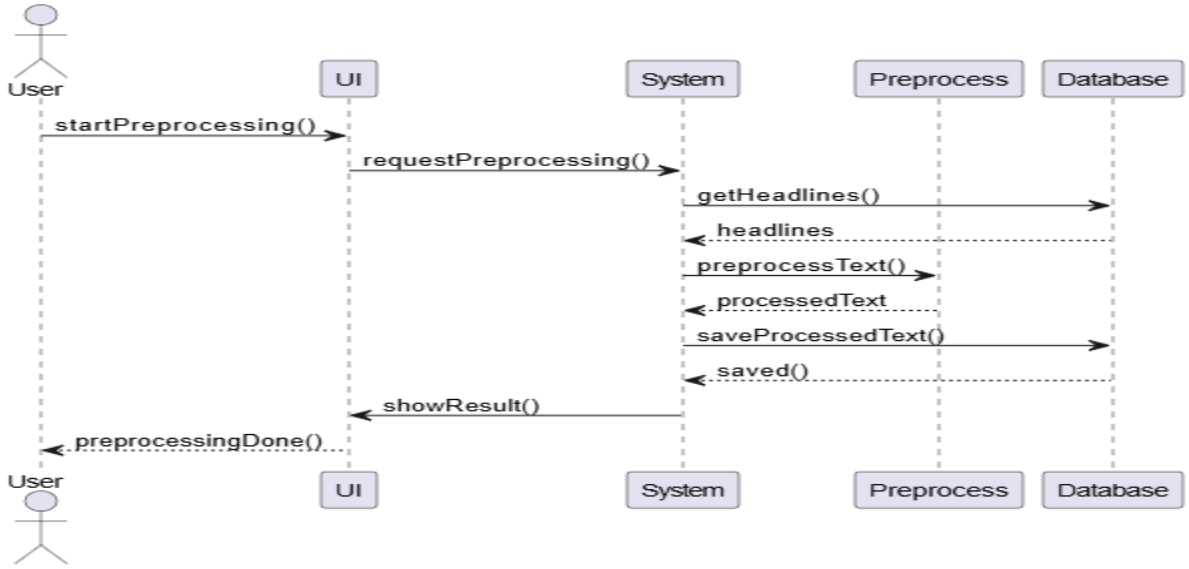


Figure 19 SD UC-5 Text Preprocessing

SD UC-6 Feature Extraction

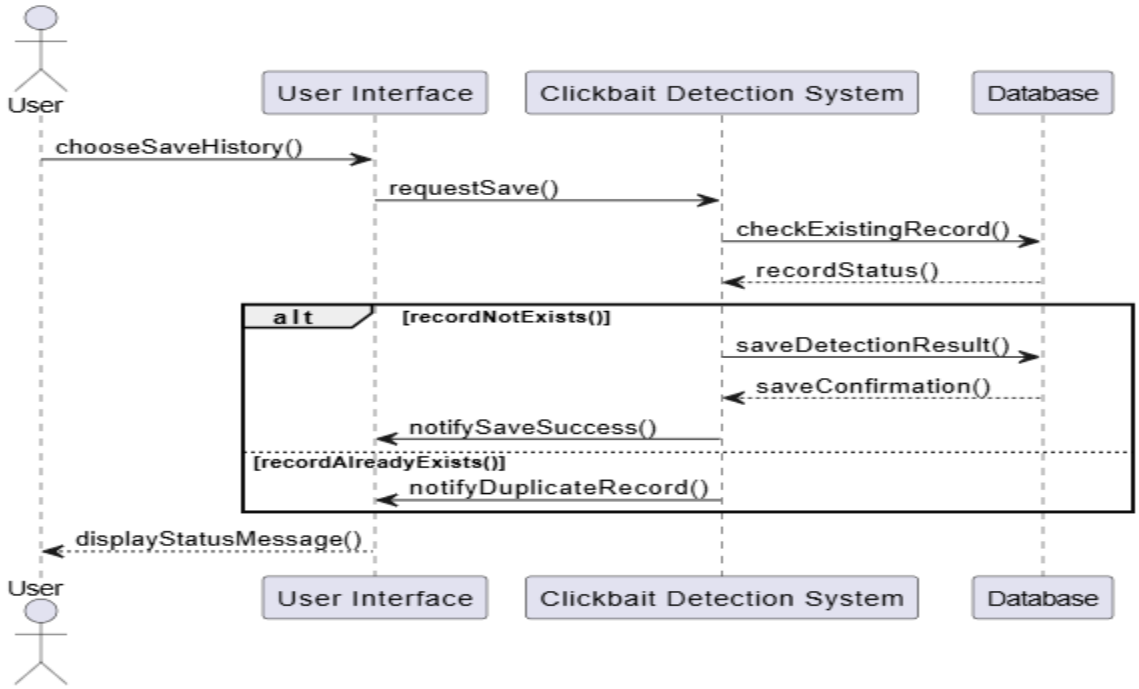


Figure 20 SD UC-6 Feature Extraction

SD UC-7 Model Training

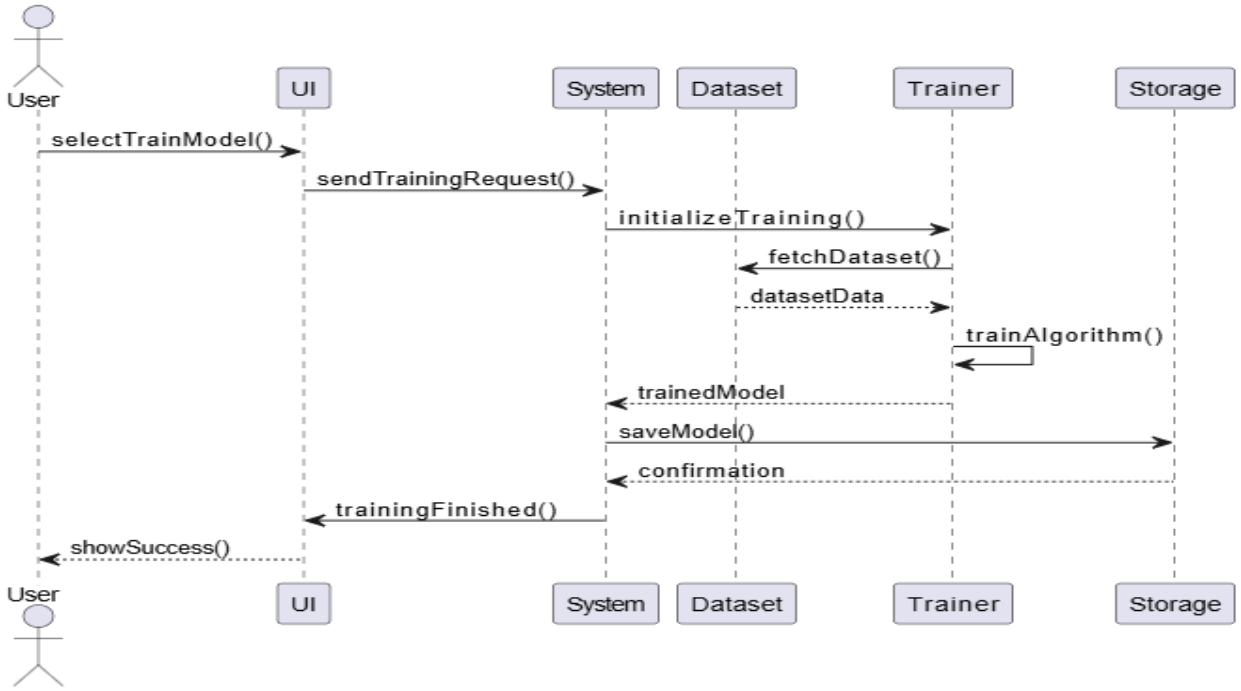


Figure 21 SD UC-7 Model Training

SD UC-8 Clickbait Prediction

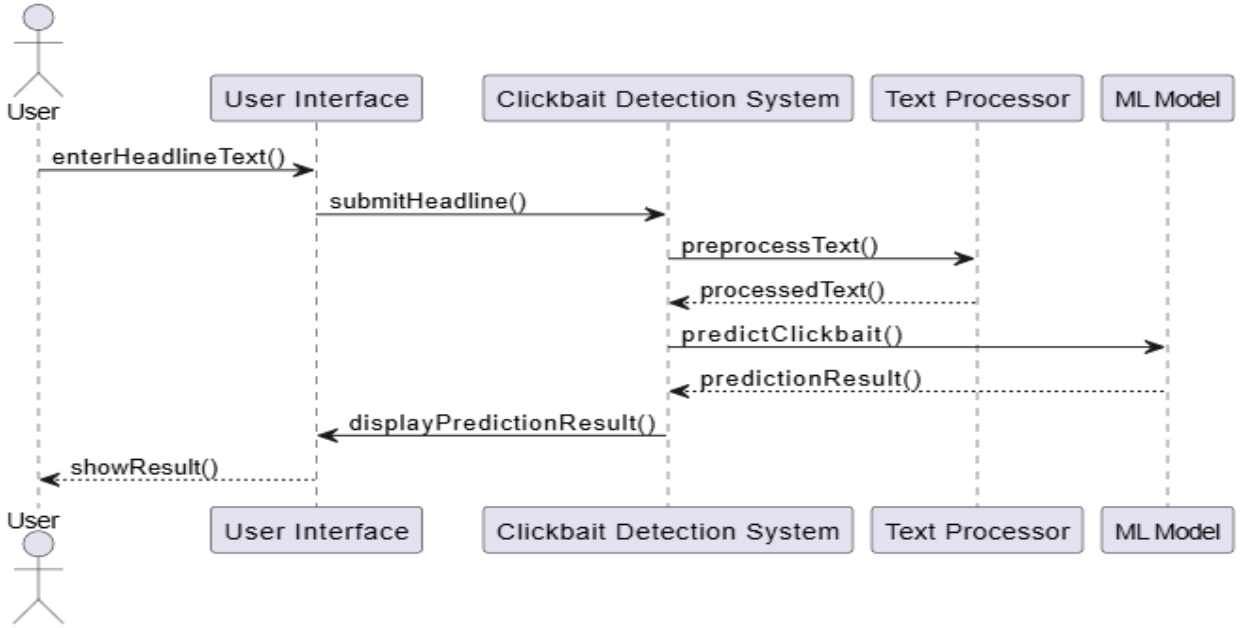


Figure 22 SD UC-8 Clickbait Prediction

SD UC-9 Explanation Generation

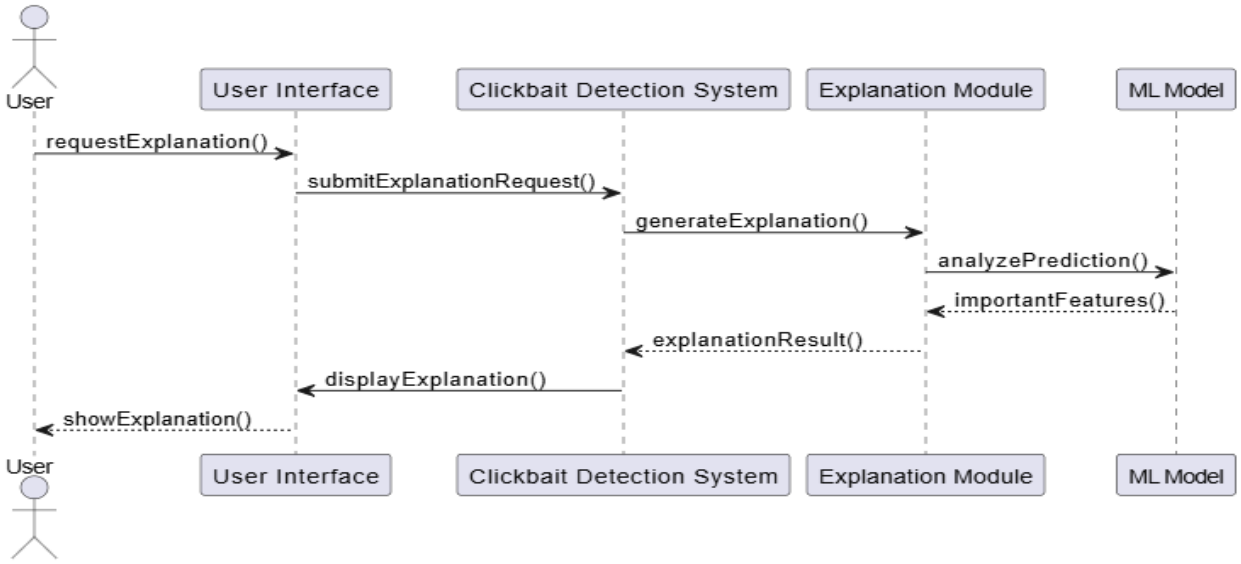


Figure 23 SD UC-9 Explanation Generation

SD UC-10 Save Detection History

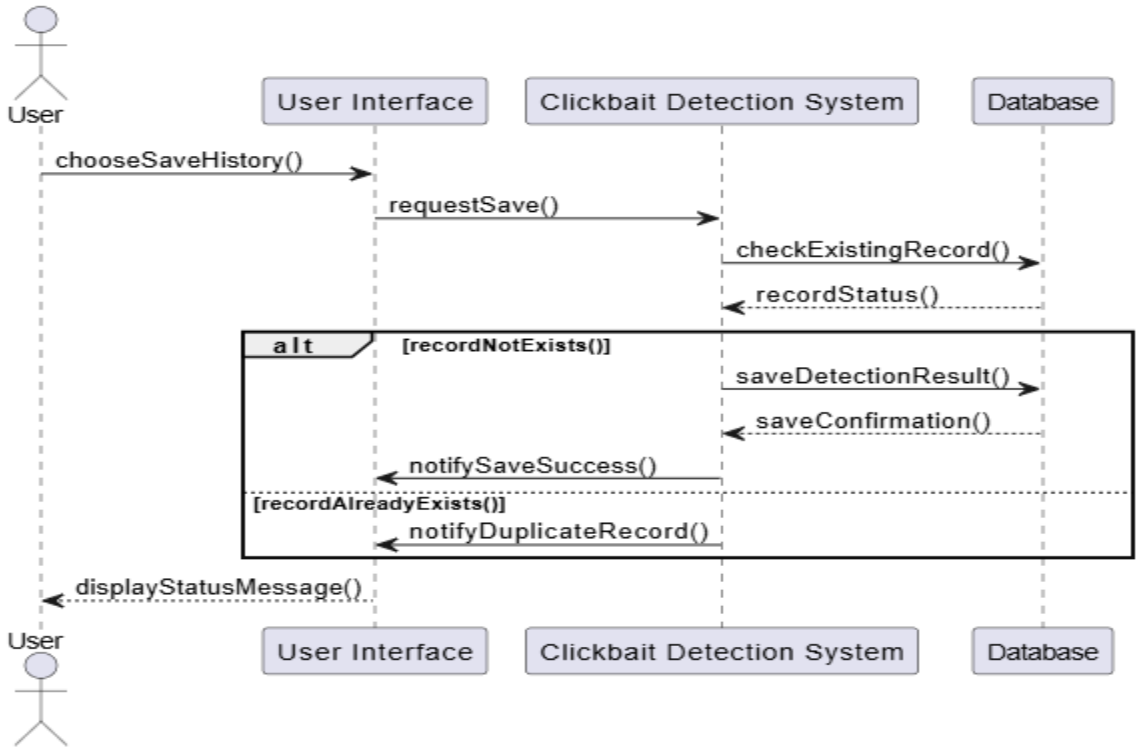


Figure 24 SD UC-10 Save Detection History

SD UC-11 View Detection History

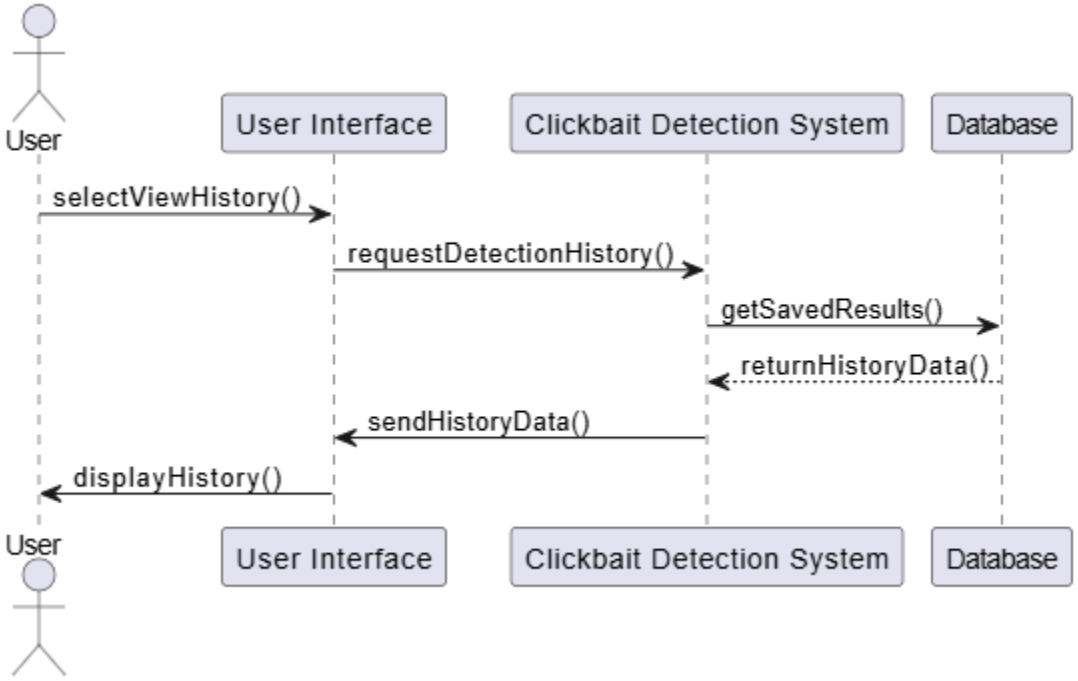


Figure 25 SD UC-11 View Detection History

Activity Diagram

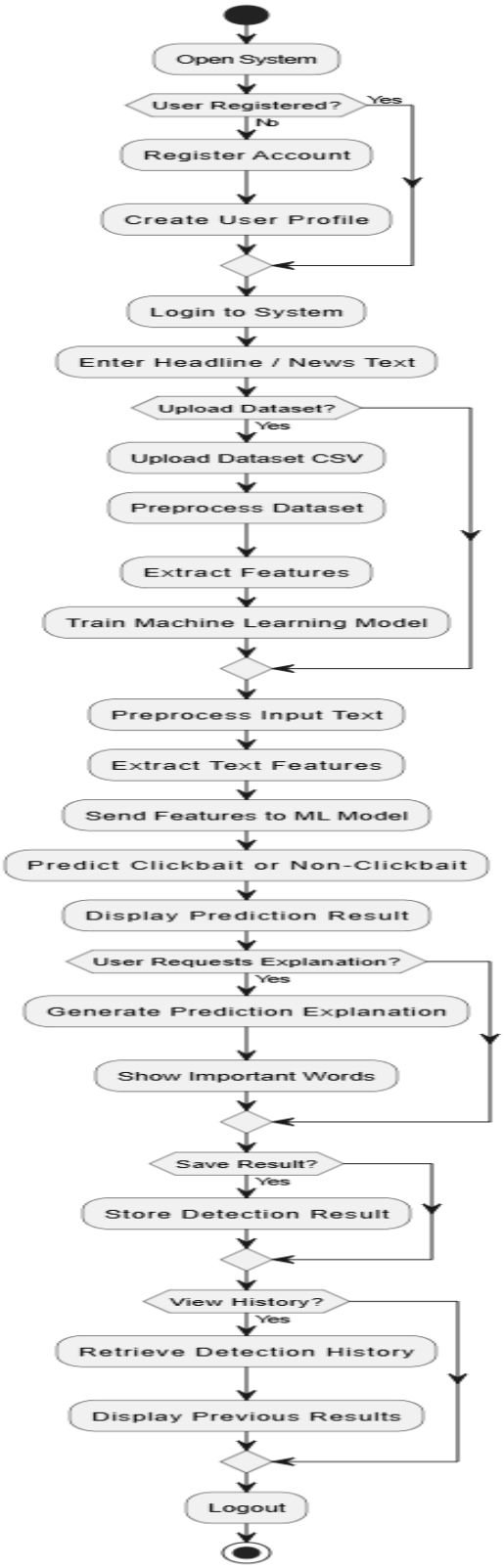


Figure 26 Activity Diagram

Class Diagram

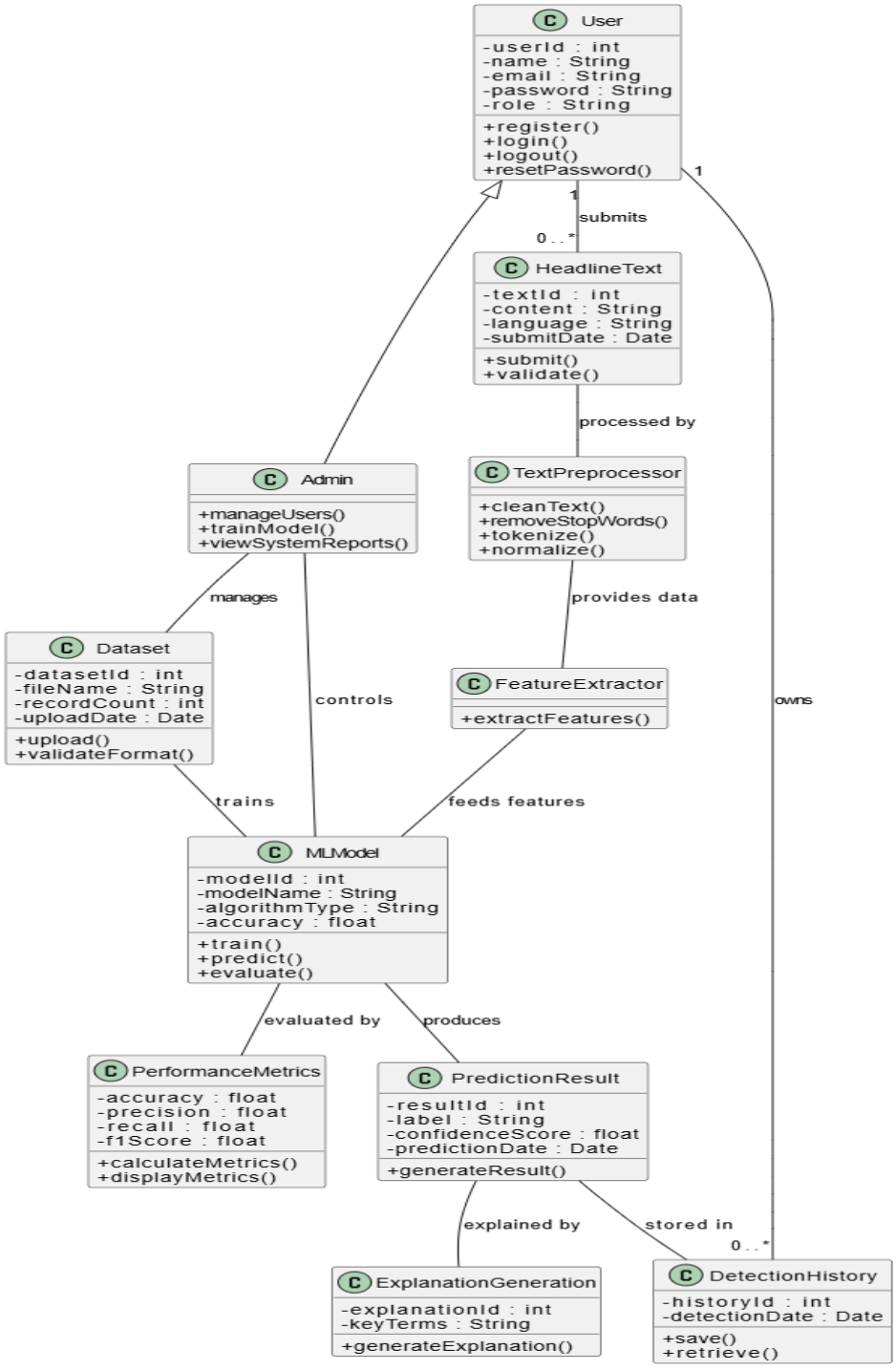


Figure 27 Class Diagram

Chapter No 4

Software Testing

Introduction

In this document we will write Acceptance test to test our project.

Acceptance Test for UC-1 User Registration

Test ID	TC-1	
Description	To test that the Clickbait Detection System lets a new consumer create an account efficiently and successfully handles incorrect or invalid registration information.	
Setup	1. Ensure the Clickbait Detection System is running in the browser. 2. Ensure the test email is not already registered in the database.	
Instruction	Valid Input	1. Open the system 2. Navigate to the User Registration page. 3. Enter the following details: 1. Name: Test User 2. Email: testuser@example.com 3. Password: Test@123 4. Confirm Password: Test@123 4. Click the Register button.
	Invalid Input	1. Enter invalid email: testuser@invalid 2. Enter mismatched passwords: Test@123 and Test@476 3. Enter already registered email: existing@example.com 4. Click the Register button for each case.
Expected Result	Valid Input	• System validates all fields. • New user account is created in the database. • System presentations message: “Registration Successful”. • User is redirected to the Login Page.
	Invalid Input	1. For invalid e-mail: System displays “Invalid e-mail format”. 2. For mismatched passwords: System presentations “Password does not fit”. 3. For current email: System shows: “Email already registered”. 4. No account is created in invalid case.

Table 14 Acceptance Test for UC-1 User Registration

Acceptance Test for UC-2 User Login

Test ID	TC-2
Description	To make sure a registered consumer can log in to the Clickbait Detection System and that the system blocks invalid login attempts.
Setup	1. Ensure the Clickbait Detection System is running in the browser. 2. Make sure that valid user account is already exists in the database. • Email: testuser@example.com • Password: Test@123

Instruction	Valid Input	1. Open the system 2. Navigate to the Login page. 3. Enter the credentials as following: • Email: testuser@example.com • Password: Test@123 4. Click the Login button.
	Invalid Input	1. Enter wrong Password: Test@ew43 2. Enter e mail that isn't always even registered: gty@exg.Com 3. Let the sector of password and electronic mail empty. 4. Try to click login button over and over.
Expected Result	Valid Input	• System verifies the credentials from the database. • User is efficaciously authenticated. • System redirects the user to the Dashboard / Home web page. • A fulfillment message may be displayed (e.g., "Login Successful").
	Invalid Input	1. For wrong password: System shows: "Invalid electronic mail or password". 2. For unregistered e mail: System presentations: "Account does no longer exist". 3. For empty fields: System displays: "Please fill all fields". 4. User isn't always allowed to get admission to the system.

Table 15 Acceptance Test for UC-2 User Login

Acceptance Test for UC-3 Reset Password

Test ID	TC-3	
Description	To verify that a registered user can efficiently reset their password and that the gadget properly handles invalid reset tries.	
Setup	1. Make sure Clickbait Detection System is running. 2. Ensure that user account already exists in the database: • Email: testuser@example.com • Old Password: Test@123	
Instruction	Valid Input	1. Open the machine. 2. Navigate to reset password web page. 3. Enter the registered e mail: testuser@instance.Com. 4. Enter new password: Newpass@123 5. Confirm new password: Newpass@123 6. Click the Reset password button.
	Invalid Input	1. Enter unregistered electronic mail: unknown@example.Com 2. Enter mismatched passwords: Newpass@123 and Newpass@671

		3. Leave fields empty. 4. Click the Reset password button for each case.
Expected Result	Valid Input	• System verifies the email exists. • Password is up to date in the database. • System displays: “Password Reset Successful”. • User can log in the use of the new password.
	Invalid Input	1. For unregistered e mail: System displays: “Email not found”. 2. For mismatched passwords: System presentations: “Passwords do not match”. 3. For empty fields: machine displays: “Please fill in all required fields”. 4. Password isn't always modified in invalid cases.

Table 16 Acceptance Test for UC-3 Reset password

Acceptance Test for UC-4 User Logout

Test ID	TC-4	
Description	To validate that an authentication consumer can securely log off of the Clickbait Detection System and that the session is properly terminated.	
Setup	1. Ensure the Clickbait Detection System is running inside the browser. 2. Ensure the user is already logged into the gadget.	
Instruction	Valid Input	1. While logged in, navigate to the dashboard or home page. 2. Click at the Logout button.
	Invalid Input	1. Attempt to get entry to a covered page (e.G., Dashboard) after logging out. 2. Use the browser again button after logout to test consultation behavior.
Expected Result	Valid Input	• The system terminates the lively person consultation. • The user is redirected to the Login page. • A confirmation message such as “Logged out correctly” can be displayed.
	Invalid Input	1. After logout, the consumer must no longer be able to access blanketed pages. 2. If attempting to get admission to a secured page, the machine redirect the consumer to the Login page. 3. The preceding consultation need to now not stay active.

Table 17 Acceptance Test for UC-4 Reset password

Acceptance Test for UC-5 Input Headlines / News Text

Test ID	TC-5

Description	To make certain that a user can input and publish a headline or news textual content for clickbait evaluation, and that the device validates the enter correctly before processing it.	
Setup	1. Ensure the Clickbait Detection System is going for walks. 2. Ensure the person is logged into the machine. 3. Ensure the text input interface is offered from the dashboard.	
Instruction	Valid Input	1. Navigate to the textual content enter segment. 2. Enter a legitimate headlines or short news text, for instance: 3. "You Won't Believe What Happened Next!". 4. Click the Analyze or Submit button.
	Invalid Input	1. Leave the text enter discipline empty. 2. Enter handiest special characters (e.G., @@@@). 3. Enter extraordinarily brief meaningless text (e.g., "aa"). 4. Click the Analyze button for each case.
Expected Result	Valid Input	• The device accepts the enter text. • The text is proven correctly. • The machine forwards the textual content to the system studying version for prediction. • No validation errors are displayed.
	Invalid Input	1. If the sphere is empty: System shows: "Text field can't be empty". 2. If the input is invalid or meaningless: System displays: "Please enter legitimate headlines or news text". 3. The gadget does not proceed to prediction in invalid cases.

Table 18 Acceptance Test for UC-5 Reset password

Acceptance Test for UC-6 Upload Dataset

Test ID	TC-6	
Description	To make certain that the machine permits customers to upload a CSV dataset for schooling the system gaining knowledge of model and efficaciously handles invalid dataset uploads.	
Setup	1. Ensure the Clickbait Detection System is strolling. 2. Ensure the consumer is logged into the gadget. 3. Ensure the dataset upload interface is accessible from the dashboard. 4. Prepare a legitimate dataset record in CSV format containing headline text and labels (clickbait / non-clickbait).	
Instruction	Valid Input	1. Navigate to the Dataset Upload phase. 2. Click the Upload Dataset button. 3. Select a valid CSV report containing clickbait dataset statistics. 4. Click the Upload / Submit button.

	Invalid Input	1. Attempt to upload a record that isn't in CSV layout (e.g., .Txt or .Pdf). 2. Upload a CSV report with lacking or wrong columns. 3. Attempt to add an empty report. 4. Click the Upload button for each case.
Expected Result	Valid Input	• The machine successfully uploads the dataset file. • The dataset is stored in the gadget for similarly processing and model schooling. • A affirmation message which include "Dataset uploaded effectively" is displayed.
	Invalid Input	1. If record layout is not CSV: System displays: "Invalid record format. Please upload a CSV file". 2. If dataset structure is wrong: System shows: "Dataset layout is invalid". 3. If report is empty: System displays: "Uploaded document carries no information". 4. The dataset isn't always popular in invalid instances.

Table 19 Acceptance Test for UC-6 Reset password

Acceptance Test for UC-7 Train Machine Learning Model

Test ID	TC-7	
Description	Confirm that the gadget trains the machine studying version the use of the uploaded dataset and handles missing or invalid statistics as it should be.	
Setup	1. Ensure the Clickbait Detection System is going for walks. 2. Ensure the person is logged into the gadget. 3. Ensure a valid clickbait has already been uploaded to the device. 4. Ensure the Model Training alternative is out there from the dashboard or training interface.	
Instruction	Valid Input	1. Navigate to the Model Training segment. 2. Ensure a valid dataset is to be had inside the device. 3. Click the Train Model button. 4. Wait for the education method to complete.
	Invalid Input	1. Attempt to teach the model with out importing any dataset. 2. Attempt to train the version with an invalid or corrupted dataset. 3. Attempt to begin the schooling technique more than one instances concurrently.
Expected Result	Valid Input	• The device hundreds the uploaded dataset. • The dataset is processed and used to educate the gadget gaining knowledge of version. • The gadget completes the schooling manner correctly.

		• A confirmation message together with "Model schooling finished efficiently" is displayed.
	Invalid Input	1. If dataset isn't to be had: System presentations: "No dataset to be had for schooling". 2. If dataset is invalid: System presentations: "Dataset format isn't always legitimate for training". 3. If training is already in progress: System displays: "Model education is already strolling". 4. Training technique isn't always started.

Table 20 Acceptance Test for UC-7 Reset password

Acceptance Test for UC-8 Detection Clickbait

Test ID	TC-8	
Description	Verify that the system analyzes the entered headlines or information text using the trained system learning version and efficaciously classifies it as Clickbait or Non- Clickbait, even as additionally managing empty or invalid enter nicely.	
Setup	1. Ensure the Clickbait Detection System is strolling. 2. Ensure the user is logged into the device. 3. Ensure the device gaining knowledge of model has already been skilled effectively. 4. Ensure the textual content input interface is available from the device dashboard.	
Instruction	Valid Input	1. Navigate to the Clickbait Detection section of the gadget. 2. Enter a headline in all likelihood to be clickbait, for instance: • "You Won't Believe What Happened Next" 3. Click the Analyze / Detect Clickbait button. 4. Enter a normal informational headline, for instance: • "Government Announces New Education Policy". 5. Click the Analyze / Detect Clickbait button once more.
	Invalid Input	1. Attempt detection without entering my text. 2. Enter most effective special characters or meaningless enter (e.G., @@@###). 3. Attempt detection when the gadget studying version is not to be had.
Expected Result	Valid Input	• The device procedures the enter text using herbal language processing techniques. • The skilled system learning model analyzes the headline.

		• The device presentations the class end result: o Clickbait, or o Non-Clickbait • The result is displayed certainly at the interface along side the prediction outcome.
	Invalid Input	1. If the enter subject is empty: System displays: “Please enter a headline or news text for evaluation”. 2. If the input is invalid: System presentations the precise validation message. 3. If the version is unavailable: System presentations: “Prediction model is not to be had”. 4. No prediction is generated invalid cases.

Table 21 Acceptance Test for UC-8 Reset password

Acceptance Test for UC-9 View Prediction Result

Test ID	TC-9	
Description	Verify that the system displays the prediction end result certainly after analyzing the input headline or news text using the skilled machine studying model.	
Setup	1. Ensure the Clickbait Detection System is walking. 2. Ensure the user is logged into the system. 3. Ensure the device studying model has already been skilled. 4. Ensure a headline or information textual content has been submitted for clickbait detection.	
Instruction	Valid Input	1. Navigate to the Clickbait Detection interface. 2. Enter a legitimate headline or news text. 3. Click the Analyze / Detect Clickbait button. 4. Wait for the gadget to process the enter and generate the prediction.
	Invalid Input	1. Attempt to view the prediction result without performing clickbait detection. 2. Refresh the web page before the prediction method completes. 3. Attempt to view prediction results while the version has not been skilled.
Expected Result	Valid Input	• The system system the enter the use of the trained machine mastering model. • The gadget presentations the prediction end result on the interface. • The result without a doubt suggests whether or not the enter text is: o Clickbait, or o Non-Clickbait. • The prediction end result is displayed in a clear and readable layout for the person.

	Invalid Input	1. If detection has no longer been done: machine displays: “No prediction result to be had”. 2. If the prediction system is incomplete: System presentations a processing or ready message. 3. If the model is unavailable: System presentations: “Prediction version now not to be had”.

Table 22 Acceptance Test for UC-9 Reset password

Acceptance Test for UC-10 Explain Prediction

Test ID	TC-10	
Description	Verify that the machine presents an cause of the prediction end result, highlighting the important thing words pr elements that stimulated the category.	
Setup	1. Ensure the Clickbait Detection System is running. 2. Ensure the consumer is logged into the device. 3. Ensure the device getting to know version has been educated efficaciously. 4. Ensure a headline or information textual content has already been analyzed and a prediction end result is to be had.	
Instruction	Valid Input	1. Navigate to the prediction end result page after appearing clickbait detection. 2. Click the explain prediction alternative. 3. Wait for the gadget to generate the reason of the prediction.
	Invalid Input	1. Attempt to view clarification with out appearing clickbait detection. 2. Attempt to get entry to explanation while no prediction end result exists. 3. Attempt to generate rationalization if the prediction technique has not finished.
Expected Result	Valid Input	• The gadget analyzes the prediction output. • The system highlights critical phrases or terms that stimulated the prediction. • The clarification is displayed really to the consumer to expose why the headline became classified as Clickbait or Non-Clickbait.
	Invalid Input	1. If prediction end result isn't always to be had: System displays: “No prediction to be had to give an explanation for”. 2. If explanation cannot be generated: System shows the correct errors message. 3. The system does no longer display rationalization without a valid prediction end result.

Table 23 Acceptance Test for UC-10 Reset password

Acceptance Test for UC-11 View Model Accuracy & Performance

Test ID	TC-11	
Description	Verify that the system efficiently shows the performance metrics of the educated machine mastering model, including accuracy and different assessment measures.	
Setup	1. Ensure the Clickbait Detection System is walking. 2. Ensure the consumer is logged into the machine. 3. Ensure the machine gaining knowledge of version has been successfully educated the usage of a legitimate dataset. 4. Ensure that the model assessment metrices had been generated after the education system.	
Instruction	Valid Input	1. Navigate to the version performance or version assessment segment of the system. 2. Select the choice view version accuracy and performance. 3. Observe the performance metrics displayed by the gadget.
	Invalid Input	1. Attempt to view model accuracy earlier than the model education process has been finished. 2. Attempt to access version overall performance metrics while no assessment information is available. 3. Refresh or reload the web page for the duration of the metrics loading manner.
Expected Result	Valid Input	• The system retrieves the version assessment metrics. • The system displays the overall performance signs sincerely on the interface. • The displayed metrics may also include: o Accuracy o Precision o Recall o F1-score • The metrics are offered in a readable and structured format for the person.
	Invalid Input	1. If the version has not been trained: device displays: “Model performance records isn't always available”. 2. If evaluation metrics are lacking: System displays the perfect notification message. 3. The machine does now not show wrong or empty overall performance statistics.

Table 24 Acceptance Test for UC-11 Reset password

Acceptance Test for UC-12 Save Detection History

Test ID	TC-12

Description	Verify that the system shops clickbait detection outcomes inside the records, inclusive of the headline text, prediction end result, and detection timestamp.	
Setup	1. Ensure the Clickbait Detection System is walking. 2. Ensure the person is logged into the system. 3. Ensure the gadget learning model has been trained correctly. 4. Ensure the user has done clickbait detection on a headline or news text.	
Instruction	Valid Input	1. Navigate to the Clickbait detection interface. 2. Enter a valid headline or information textual content for evaluation. 3. Click the Analyze / Detect Clickbait button. 4. Wait for the prediction Result to appear. 5. Confirm that the gadget routinely saves the detection result in the history phase.
	Invalid Input	1. Attempt to shop detection history without appearing clickbait detection. 2. Attempt to store detection facts while the prediction procedure fails. 3. Attempt to store empty or invalid detection data.
Expected Result	Valid Input	• The gadget successfully stores the detection bring about the records database. • The saved document Includes: o Input headline or new textual content. o Prediction end result (Clickbait / Non-Clickbait). o Detection date and time. • A confirmation message which includes "Detection end result saved efficiently" can be displayed.
	Invalid Input	1. If no detection has been finished: System does no longer create a history record. 2. If detection fails: System does not keep incomplete statistics. 3. The machine shows the ideal notification indicating that the file can not be saved.

Table 25 Acceptance Test for UC-12 Reset password

Acceptance Test for UC-13 View Detection History

Test ID	TC-13
Description	Allow users to view formerly stored clickbait detection outcomes, displaying the headline, prediction end result, and detection time.
Setup	1. Ensure the Clickbait Detection System is walking. 2. Ensure the person is logged into the device. 3. Ensure that at the least one clickbait detection end result has been stored inside the system records.

Instruction	Valid Input	1. Navigate to the detection history section from the machine dashboard. 2. Select the option view detection history. 3. Observe the list of formerly stored detection information.
	Invalid Input	1. Attempt to view detection history whilst no detection information exists in the machine. 2. Attempt to access detection history without logging into the machine. 3. Refresh or reload the records web page for the duration of the loading manner.
Expected Result	Valid Input	• The machine retrieves the stored detection history from the database. • The device presentations the previous detection statistics in a established layout. • Each file have to consist of: o The analyzed headline or news textual content. o The prediction end result (Clickbait / Non-Clickbait). o The date and time of detection.
	Invalid Input	1. If no records information exsist: System displays: “No detection history available”. 2. If the consumer isn't always logged in: System redirects the person to the login web page. 3. The gadget should now not displayed empty or incorrect history records.

Table 26 Acceptance Test for UC-13 Reset password

Chapter No 5

Conclusion and Future Enhancement

Introduction

In this chapter we are able to see how we are able to enhance the app to make it more and more higher to serve mankind.

Summary and Conclusion

The area of expertise of this challenge is Clickbait Detection System the usage of Machine Learning to come across misleading and sensational headlines in on line platforms. The device analyzes the text of a headline and classifies it as clickbait or non-clickbait to assist users pick out exaggerated or unverifiable content material. Users can either write headlines or upload datasets in order to train the gadget gaining knowledge of version. Post training, the model predicts and offers a concise reasoning with the aid of figuring out essential words that helped in classifying. It also shows version accuracy and performance that can assist investigate the effectiveness of a given model.

Moreover, detection results are saved in a records segment to permit customers to test the previous analyses. Using larger datasets and extra state-of-the-art device gaining knowledge of techniques inside the future can beautify detection accuracy.

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